

Practical Data Science: Wrangling Data and Answering Questions

Nick Eubank

What is Data Science?

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2. What (empirically) is Data Science?

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 - Data Science is an *applied* discipline!

What (in theory) should Data Science be?

Study of how to **solve problems** by **answering questions** using **quantitative data**.

- Problem-first approach.
 - Data Science is an *applied* discipline!
- The tool you use should be dictated by the question you seek to answer

What (empirically) is Data Science?

How did Data Science become a thing?

Over the past several decades:

1. Availability of data ↑
2. Computational power ↑

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⇒ Huge proliferation and increase in sophistication of computational methods

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 - Economics
 - Political science
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⇒ Development of new tools occurred *within* each silo.

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Very little cross-pollination across silos

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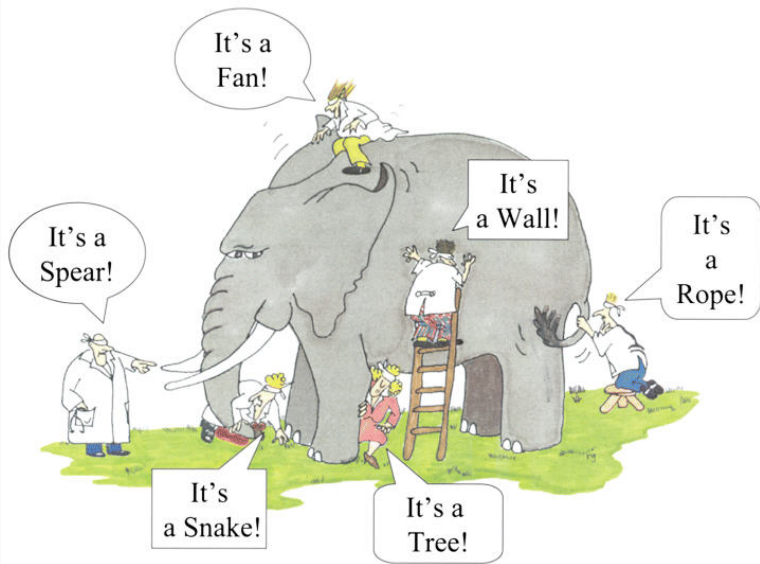
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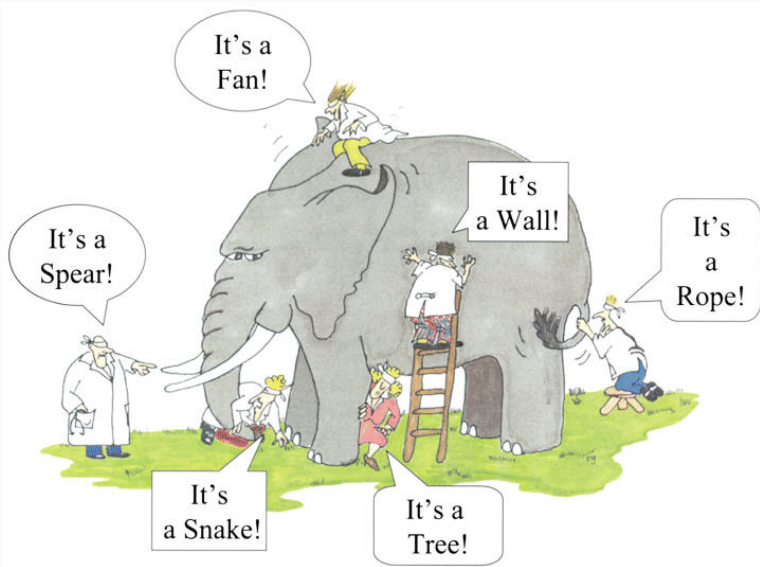
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 - CS likes to classify things and make predictions, don't care how model works
 - Social scientists like to make causal statements, don't care about predictive power





⇒ This is where we are *now*.

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→ Recognize the elephant

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 - That's why we have faculty from many fields.

Areas of Data Science

Software Engineering DS

Data Analysis DS

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- Recommendation engines
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Data Analysis DS

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Nearly all data scientists will use some of both sets of skills. In this class, we will be focused on the **Data Analysis** flavor of Data Science.

This Class

“The Python Class”

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- *Data Science Python*

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 - numpy, pandas, scikit-learn, statsmodels, dask

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All through hands-on experience.

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⇒ Emphasis on generalizable data science skills

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- Find and organize data *on your own*.

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And why not:

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- R (Tidy-Verse)
- R (Base-R)

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Excellent for tabular data

- R (Base-R)

Good for tabular, network, geospatial, some text and ML

?

Why Python?

Python:

- Tabular data
- Network data
- Geospatial data
- Natural Language Processing (NLP)
- Neural Networks
- Using with Cloud Compute
- Big Data
- Large Language Models
- All Machine Learning
- ...

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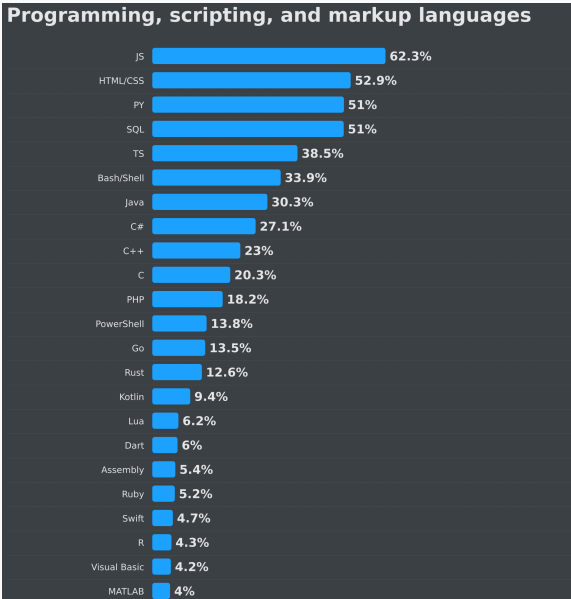
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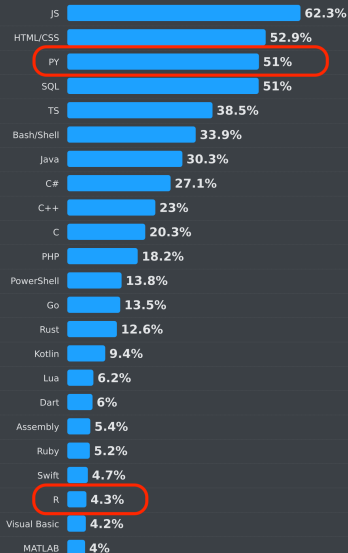
- 90s and 2000s (even 2010s): most social scientists used Stata or R, developed for Stata and R.
- 2010s and 2020s: Businesses, computer scientists and software engineers moved to data science.
 - Wanted a fully-featured, general purpose language (ideally one they know).

Stack Overflow Developer Survey 2024

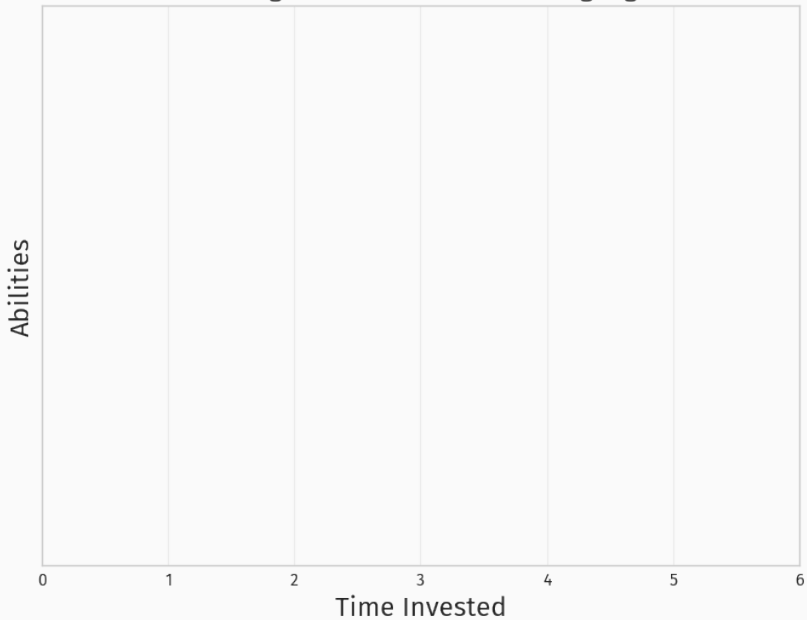


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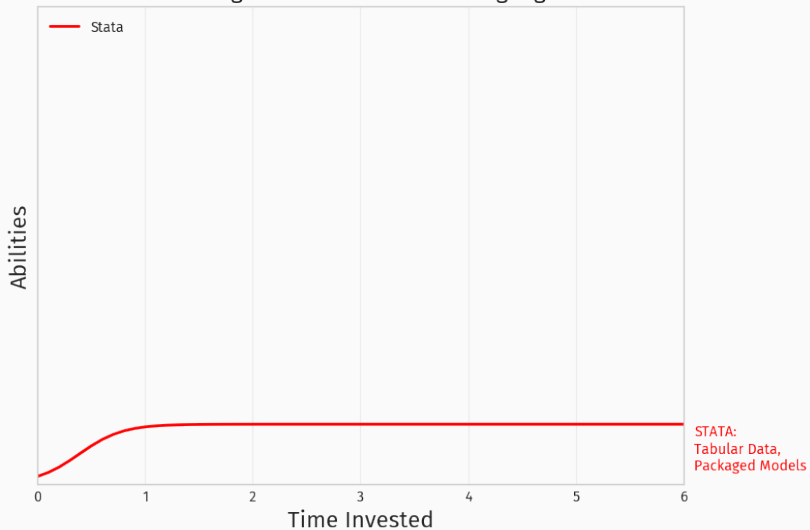
Programming, scripting, and markup languages



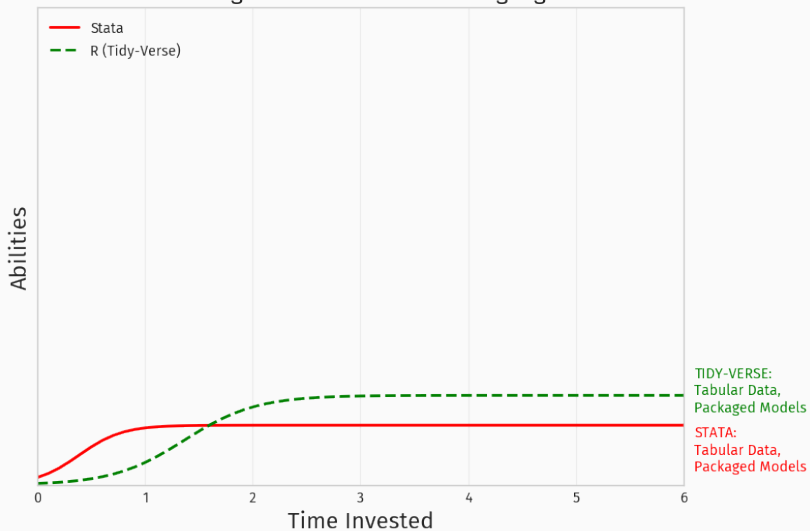
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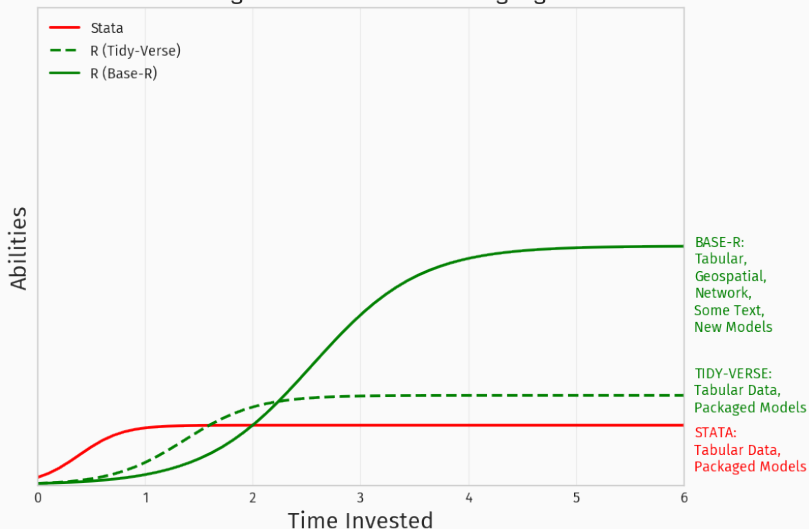
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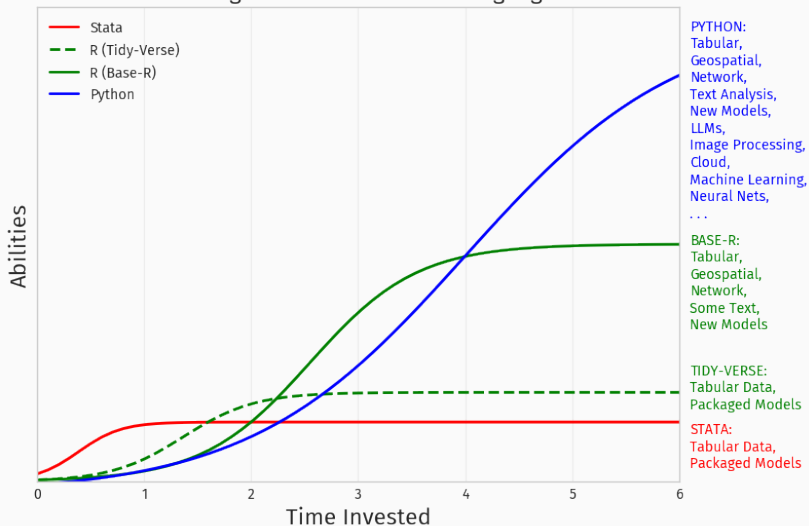
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 - (Also easier ways in Python!)
- “This isn’t what I want to learn.”
 - Especially for the first 4 weeks.

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- and more.

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About Me

I am a social scientist

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- Testing theories about how social networks shape political behavior using cell-phone meta-data to map social networks of entire countries (Zambia and Venezuela).
- Studying whether political elites in the US South turned to using incarceration to prevent black voters from exercising political influence after the Voting Rights Act removed their ability to use Jim Crow restrictions.

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If this sounds like your “flavor” of data science, I'm happy to talk about career options in this domain.

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www.practicaldatascience.org